

Incident Report

Failed Login Detection

SOC/SIEM Detection Lab: Wazuh, Windows Security Logs, and Hyper-V

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Scenario	Failed login / authentication failure detection
Classification	Portfolio lab documentation - controlled simulation

Report Purpose

This report documents a controlled authentication-monitoring scenario in a home SOC/SIEM lab. The objective was to generate failed logon activity on a monitored Windows endpoint, validate local Windows Security logging, confirm Wazuh alert ingestion, and document cleanup of the test account.

1. Executive Summary

A controlled failed-login scenario was performed on WIN11-ENDPOINT, a Windows 11 endpoint monitored by Wazuh. A temporary local test account named failedloginuser was created, followed by multiple intentional incorrect-password attempts. Windows Security logs recorded Event ID 4625, and Wazuh generated corresponding logon failure alerts for the endpoint.

This scenario demonstrated authentication monitoring, Windows Security event analysis, SIEM alert validation, and post-test cleanup. The test user was deleted after evidence collection, and account removal was verified from PowerShell.

Key Findings

- Windows recorded failed logon events using Security Event ID 4625.
- Wazuh ingested and displayed three logon failure alerts for WIN11-ENDPOINT.
- The failed target account was failedloginuser.
- The rule description in Wazuh was Logon Failure - Unknown user or bad password.
- The temporary test account was removed and cleanup was verified.

2. Environment

Hypervisor	Microsoft Hyper-V
SIEM Platform	Wazuh
SIEM Server	Ubuntu Server / Wazuh-SIEM
Wazuh Manager IP	192.168.56.10
Monitored Endpoint	WIN11-ENDPOINT
Endpoint IP	192.168.56.20
Endpoint OS	Windows 11 Enterprise Evaluation
Primary Log Source	Windows Security Logs
Relevant Event ID	4625 - An account failed to log on

3. Scenario Objective

The objective was to simulate a simple authentication failure pattern that a SOC analyst may investigate during routine monitoring. Multiple failed login attempts were generated against a known local test account. The goal was to confirm that the endpoint logged the behavior locally and that Wazuh received and surfaced the activity for investigation.

Simulated Activity

The following controlled activity was performed:

- A temporary test account named failedloginuser was created on WIN11-ENDPOINT.
- Incorrect-password attempts were generated using runas to simulate failed local logons.
- Windows Security Event ID 4625 was validated locally.
- Wazuh Threat Hunting was used to find and review failed logon alerts.
- The test account was deleted and cleanup was verified.

Commands Used

```
net user failedloginuser [known lab password] /add
runas /user:WIN11-ENDPOINT\failedloginuser cmd
# Incorrect passwords were entered intentionally to generate failed logons.
net user failedloginuser /delete
net user failedloginuser
```

The known lab password is intentionally omitted from this report. No real credentials were used.

4. Evidence Summary

Evidence Item	Source	Key Fields / Finding
Failed logon generated	Windows Security Log	Event ID 4625; target account: failedloginuser; failure reason: unknown user name or bad password.
Wazuh alert overview	Wazuh Threat Hunting	Three logon failure events were visible for WIN11-ENDPOINT during the test window.
Wazuh event details	Wazuh Document Details	Event ID 4625; targetUserName failedloginuser; channel Security; severity AUDIT_FAILURE.
Cleanup verification	Windows PowerShell	failedloginuser was deleted and confirmed no longer present.

5. Detection and Analysis

Local Windows Security Evidence

Windows Security logs recorded Event ID 4625, indicating that an account failed to log on. The event included the test account failedloginuser, a failure reason of unknown user name or bad password, and the endpoint WIN11-ENDPOINT as the workstation/computer involved in the logon attempt.

Administrator: Windows PowerShell

```

TimeCreated : 5/18/2026 10:42:24 AM
Id : 4625
ProviderName : Microsoft-Windows-Security-Auditing
Message : An account failed to log on.

Subject:
  Security ID: S-1-5-21-417507766-3430154192-941633884-1001
  Account Name: labuser
  Account Domain: WIN11-ENDPOINT
  Logon ID: 0xCA306

Logon Type: 2

Account For Which Logon Failed:
  Security ID: S-1-0-0
  Account Name: failedloginuser
  Account Domain: WIN11-ENDPOINT

Failure Information:
  Failure Reason: Unknown user name or bad password.
  Status: 0xC000006D
  Sub Status: 0xC000006A

Process Information:
  Caller Process ID: 0x880
  Caller Process Name: C:\Windows\System32\svchost.exe

Network Information:
  Workstation Name: WIN11-ENDPOINT
  Source Network Address: ::1
  Source Port: 0

Detailed Authentication Information:
  Logon Process: seclogo
  Authentication Package: Negotiate
  Transited Services: -
  Package Name (NTLM only): -
  Key Length: 0

This event is generated when a logon request fails. It is generated on the computer where access was attempted.

The Subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.

The Logon Type field indicates the kind of logon that was requested. The most common types are 2 (interactive) and 3 (network).

```

Figure 1. Local Windows Security Event ID 4625 showing failed logon activity for failedloginuser.

Wazuh Alert Overview

Wazuh Threat Hunting displayed three failed logon alerts for WIN11-ENDPOINT during the test window. The rule description showed Logon Failure - Unknown user or bad password, confirming that Wazuh ingested the Windows Security events and generated alert records.

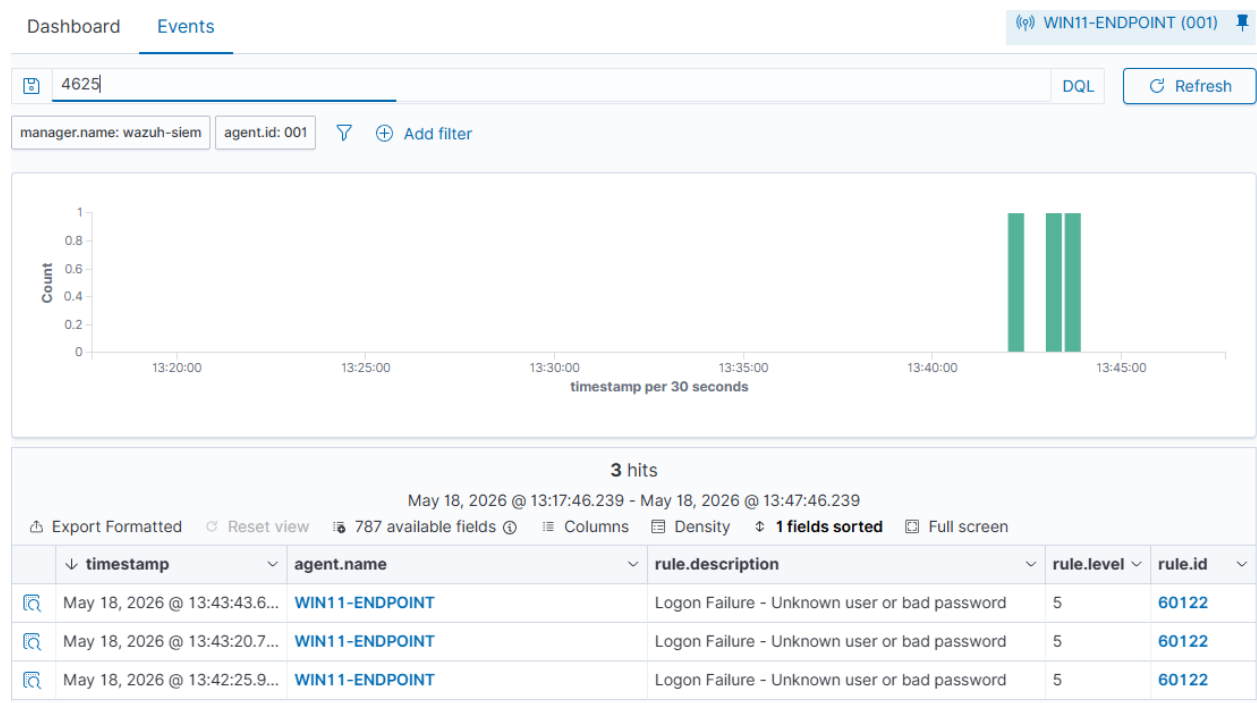


Figure 2. Wazuh Threat Hunting overview showing three failed logon events for WIN11-ENDPOINT.

Wazuh Event Details

The Wazuh document details confirmed key event fields, including the endpoint name, target user, Security channel, Event ID 4625, audit failure severity, and rule description.

data.win.eventdata.targetDomainName	WIN11-ENDPOINT
data.win.eventdata.targetUserName	failedloginuser
data.win.eventdata.targetUserSid	S-1-0-0
data.win.eventdata.workstationName	WIN11-ENDPOINT
data.win.system.channel	Security
data.win.system.computer	WIN11-ENDPOINT
data.win.system.eventID	4625
data.win.system.eventRecordID	30351
data.win.system.keywords	0x8010000000000000
data.win.system.level	0
data.win.system.message	"An account failed to log on. Subject: Security ID: S-1-5-21-417507766-3430154192-941633884-1001 Account Name: 1
data.win.system.opcode	0
data.win.system.processID	880
data.win.system.providerGuid	{54849625-5478-4994-a5ba-3e3b0328c30d}

Figure 3. Wazuh event details showing target username, Security channel, Event ID 4625, and failed logon message.

t decoder.name	windows_eventchannel
t id	1779126223.209698
t input.type	log
t location	EventChannel
t manager.name	wazuh-siem
t rule.description	Logon Failure - Unknown user or bad password

Figure 4. Wazuh event details showing manager name and rule description for the logon failure alert.

6. Investigation Steps

1. Generated failed logon attempts using the temporary local account failedloginuser.
2. Queried Windows Security logs for Event ID 4625 and filtered for failedloginuser.
3. Confirmed the failure reason as unknown user name or bad password.
4. Opened Wazuh Threat Hunting and searched for Event ID 4625.
5. Confirmed Wazuh alerts for WIN11-ENDPOINT with the rule description Logon Failure - Unknown user or bad password.
6. Reviewed Wazuh document details to validate the target username, event ID, source channel, and alert rule.
7. Deleted the temporary test user after evidence collection.
8. Verified that the test account no longer existed on the endpoint.

7. Potential Impact

In a production environment, repeated failed logon activity may indicate password guessing, brute-force attempts, misconfigured services, stale credentials, or unauthorized access attempts. Investigating failed logons helps determine whether the activity is benign, user error, automation-related, or part of an attack pattern.

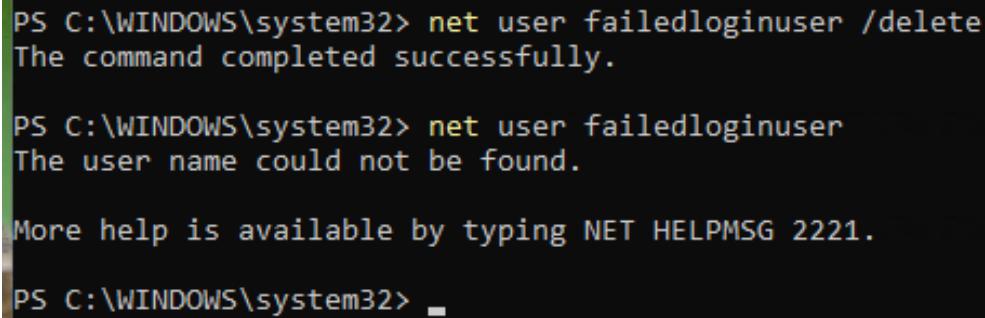
8. MITRE ATT&CK Mapping

Technique	Tactic	Relevance to Scenario
T1110 - Brute Force	Credential Access	Multiple failed logon attempts can indicate brute-force-style authentication behavior.
T1110.001 - Password Guessing	Credential Access	The lab simulated repeated incorrect password attempts against a known local user account.

This mapping is used as a lab-oriented reference. The activity was controlled and benign, but the observed behavior is relevant to authentication attack monitoring.

9. Containment and Remediation

After the investigation was completed, the temporary test account was deleted from WIN11-ENDPOINT. Cleanup was verified by querying the account and confirming that Windows returned that the username could not be found.



```
PS C:\WINDOWS\system32> net user failedloginuser /delete
The command completed successfully.

PS C:\WINDOWS\system32> net user failedloginuser
The user name could not be found.

More help is available by typing NET HELPMSG 2221.

PS C:\WINDOWS\system32> _
```

Figure 5. PowerShell verification showing failedloginuser deleted and no longer present.

10. Lessons Learned

- Windows Security Event ID 4625 is a key event for failed logon monitoring.
- Wazuh can ingest and surface Windows Security audit failures for endpoint authentication activity.
- Failed logon investigation should include user, workstation, failure reason, logon type, timestamp, and event volume.
- Clean lab procedures include removing temporary test accounts after completing evidence collection.

11. Skills Demonstrated

- Windows Security log analysis
- Event ID 4625 investigation
- Wazuh Threat Hunting workflow
- Authentication monitoring
- PowerShell-based event validation
- SIEM evidence review
- Basic incident documentation and cleanup verification

12. Conclusion

The failed login detection scenario successfully demonstrated local Windows event validation and Wazuh SIEM alert review. The test generated multiple Event ID 4625 records, Wazuh surfaced the corresponding failed logon alerts, and the temporary account was removed after the investigation. This scenario strengthens the SOC/SIEM lab by adding authentication-monitoring evidence to the existing endpoint detection portfolio.

Appendix A: Evidence Screenshots

The following screenshots were used as supporting evidence in this report:

- 73-Windows-Failed-Logon-4625-Local.png
- 74-Wazuh-Failed-Logon-Event-Overview.png
- 75-Wazuh-Failed-Logon-Event-Details.png
- 76-Failed-Login-Test-User-Cleanup-Verified.png